

SSL/TLS Certificate Process

End-to-End Guide: CSR → Signed Certificate → ADF REST Linked Service

Coverage	Details
Platform	Windows PowerShell
Authentication	Certificate-based (Service Principal)
Target Service	Microsoft Graph API via ADF REST Linked Service
CA Flow	You generate CSR → Cyber Security team signs → You install
Storage	Direct Base64 embed in ADF (no Key Vault required)

KEY CONCEPTS BEFORE YOU START

Term	Full Form	What It Does
CSR	Certificate Signing Request	Your application to the CA — contains your public key + identity
CA	Certificate Authority	Trusted third party (your Cyber Security team) that signs your CSR
PFX	Personal Information Exchange	File that bundles private key + signed certificate together
CER	Certificate File	Public certificate only — safe to share, upload to Azure AD
Base64	Binary-to-Text Encoding	How ADF accepts the PFX without Key Vault — as a text string
OID	Object Identifier	Number that defines what the certificate is allowed to do

■ The certificate handles AUTHENTICATION (who you are). API Permissions on the App Registration handle AUTHORIZATION (what you can do). These are completely separate — replacing a certificate never affects permissions.

OVERALL PROCESS FLOW

#	Who	Action	Output
1	You	Create request.inf config file	request.inf
2	You	Generate CSR using certreq	request.csr + Private Key
3	You → Cyber Team	Send request.csr to Cyber Security team	—
4	Cyber Team → You	Cyber team signs and returns signed cert	mycert.cer
5	You	Install signed cert with certreq -accept	Cert in Windows Store
6	You	Export PFX (private key + cert bundled)	myapp.pfx
7	You	Export CER (public cert only)	myapp.cer

8	You	Convert PFX to Base64 string	cert_base64.txt
9	You	Upload CER to Azure AD App Registration	Thumbprint noted
10	You	Paste Base64 into ADF REST Linked Service	Linked Service ready
11	You	Test Connection in ADF	✓ Connected

STEP 1 — GENERATE CSR & PRIVATE KEY

Run these PowerShell commands on the machine where ADF will authenticate. The private key is automatically stored in the Windows Certificate Store and never needs to be shared.

Create Folder

```
# Create working folder
New-Item -ItemType Directory -Path 'C:\certs' -Force
```

Create request.inf Config File

```
$inf = @"
[Version]
Signature = "$Windows NT$"

[NewRequest]
Subject = "CN=ADF-GraphAPI, O=Acme Corp, C=IN"
KeySpec   = 1           ; 1 = AT_KEYEXCHANGE (sign + encrypt)
KeyLength = 2048        ; RSA key size in bits
Exportable = TRUE       ; Must be TRUE to export PFX later
MachineKeySet = TRUE    ; Store in LocalMachine (not current user)
RequestType = PKCS10    ; Standard CSR format expected by all CAs

[EnhancedKeyUsageExtension]
OID = 1.3.6.1.5.5.7.3.2 ; Client Authentication (for Azure AD / ADF)
"@
$inf | Out-File -FilePath 'C:\certs\request.inf' -Encoding ASCII
```

request.inf Field Reference

Field	Value	Meaning
Signature	\$Windows NT\$	Magic string — always exactly this, never change
Subject CN	ADF-GraphAPI	App name — for Azure AD use app name, not domain
KeySpec	1	1 = key can sign AND encrypt (required for TLS/auth)
KeyLength	2048	RSA key strength — 2048 is standard, 4096 is stronger
Exportable	TRUE	MUST be TRUE — lets you export PFX for ADF later
MachineKeySet	TRUE	Stores key at machine level — survives user logoff
RequestType	PKCS10	Standard CSR format — what your Cyber team expects
OID	1.3.6.1.5.5.7.3.2	Client Authentication — correct OID for Azure AD

Generate the CSR

```
# This creates request.csr AND stores private key in Windows cert store
certreq -new 'C:\certs\request.inf' 'C:\certs\request.csr'
```

■ After running this, send C:\certs\request.csr to your Cyber Security team. The private key stays on your machine automatically — never share it.

STEP 2 — INSTALL SIGNED CERTIFICATE FROM CYBER TEAM

Your Cyber Security team will return a signed .cer file. Run the following on the SAME machine that generated the CSR — this links the signed cert with your private key.

```
# Install signed certificate — links with your existing private key
certreq -accept 'C:\certs\mycert.cer'
```

Verify Installation

```
# Confirm cert is installed and private key is attached
Get-ChildItem -Path 'Cert:\LocalMachine\My' |
  Where-Object { $_.Subject -like '*ADF-GraphAPI*' } |
  Select-Object Subject, Thumbprint, NotAfter, HasPrivateKey
```

■ *HasPrivateKey must be True. If it shows False, you ran certreq -accept on the wrong machine. It must be the same machine that ran certreq -new in Step 1.*

STEP 3 — EXPORT PFX AND CER

Export two files — the PFX (private key + cert, for ADF) and the CER (public cert only, for Azure AD App Registration).

Export PFX — Private Key + Certificate

```
# Get your certificate from the store
$cert = Get-ChildItem -Path 'Cert:\LocalMachine\My' |
  Where-Object { $_.Subject -like '*ADF-GraphAPI*' }

# Set a strong password
$pwd = ConvertTo-SecureString -String 'YourPassword123' -Force -AsPlainText

# Export PFX — contains private key + certificate bundled
Export-PfxCertificate `
  -Cert $cert `
  -FilePath 'C:\certs\myapp.pfx' `
  -Password $pwd
```

Export CER — Public Certificate Only

```
# Export public cert only — this goes to Azure AD App Registration
Export-Certificate `
  -Cert $cert `
  -FilePath 'C:\certs\myapp.cer' `
  -Type CERT
```

File	Contains	Used For
myapp.pfx	Private Key + Certificate	ADF REST Linked Service (Base64 encoded)
myapp.cer	Public Certificate only	Azure AD App Registration upload

STEP 4 — CONVERT PFX TO BASE64 STRING

ADF accepts the certificate directly as a Base64 encoded string — no Key Vault required. Convert your PFX to Base64 and save it to a text file.

```
# Read PFX bytes and convert to Base64
$pfXBytes = [System.IO.File]::ReadAllBytes('C:\certs\myapp.pfx')
$base64Cert = [System.Convert]::ToBase64String($pfXBytes)

# Save to text file – you will paste this into ADF
$base64Cert | Out-File -FilePath 'C:\certs\cert_base64.txt' -Encoding ASCII

Write-Host 'Done. Open C:\certs\cert_base64.txt and copy the full content.'
```

■ *Open cert_base64.txt and copy the entire content. It will be a very long single line of text — this is normal. You will paste it into ADF in Step 7.*

STEP 5 — REGISTER APP IN AZURE AD & UPLOAD CER

Create or use an existing App Registration in Azure AD (Entra ID), upload the public certificate, and grant the required API permissions.

#	Where	Action
1	Azure Portal → Entra ID → App Registrations	Click New Registration, name it ADF-GraphAPI, click Register
2	Overview page	Note down Application (Client) ID and Directory (Tenant) ID
3	Certificates & Secrets → Certificates	Click Upload Certificate → upload myapp.cer → note Thumbprint
4	API Permissions → Add Permission	Select Microsoft Graph → Application Permissions
5	API Permissions	Add required permissions (see table below)
6	API Permissions	Click Grant Admin Consent → confirm Yes

Recommended API Permissions

Permission	Type	Use Case
User.Read.All	Application	Read all users in tenant
Sites.Read.All	Application	Read all SharePoint sites
Sites.Selected	Application	Read specific SharePoint sites only (more secure)
Files.Read.All	Application	Read files and documents
Sites.ReadWrite.All	Application	Read and write all SharePoint sites

■ *Sites.Selected is recommended over Sites.Read.All — it grants access to only specific sites rather than all sites in the tenant. Note: Permissions belong to the App Registration (Client ID), not to the certificate. Replacing a certificate never changes permissions.*

STEP 6 — GRANT SITE ACCESS FOR SITES.SELECTED (IF USED)

If you chose Sites.Selected permission, a SharePoint Site Admin must explicitly grant your app access to specific sites. Skip this step if using Sites.Read.All.

```
# Install PnP PowerShell module
Install-Module PnP.PowerShell -Force

# Connect to your SharePoint site
Connect-PnPOnline -Url 'https://yourtenant.sharepoint.com/sites/YourSite' `
-Interactive

# Grant your ADF app access to this specific site
Grant-PnPazureADAppSitePermission `
-AppId '<YOUR-CLIENT-ID>' `
-DisplayName 'ADF-GraphAPI' `
-Site 'https://yourtenant.sharepoint.com/sites/YourSite' `
-Permissions Read
```

STEP 7 — CREATE REST LINKED SERVICE IN ADF

In ADF Studio, create a new REST Linked Service and embed the Base64 certificate directly — no Key Vault needed.

ADF Studio UI — Field Values

Field	Value
Linked Service Type	REST
Base URL	https://graph.microsoft.com/v1.0/
Authentication Type	Service Principal
Tenant ID	Your Directory (Tenant) ID from Step 5
Service Principal ID	Your Application (Client) ID from Step 5
Credential Type	Service Principal Certificate
Certificate	Paste full content of cert_base64.txt here
Certificate Password	YourPassword123 (same password used in Export-PfxCertificate)

ADF JSON Definition

```
{
  "name": "LS_GraphAPI_DirectCert",
  "type": "Microsoft.DataFactory/factories/linkedservices",
  "properties": {
    "type": "RestService",
    "typeProperties": {
      "url": "https://graph.microsoft.com/v1.0/",
      "authenticationType": "ServicePrincipal",
      "servicePrincipalId": "<YOUR-CLIENT-ID>",
      "tenant": "<YOUR-TENANT-ID>",
```

```
"aadResourceId": "https://graph.microsoft.com",
"servicePrincipalCredentialType": "ServicePrincipalCert",
"servicePrincipalEmbeddedCert": {
  "type": "SecureString",
  "value": "<PASTE-BASE64-STRING-HERE>"
},
"servicePrincipalEmbeddedCertPassword": {
  "type": "SecureString",
  "value": "YourPassword123"
}
}
```


STEP 8 — TEST CONNECTION & VERIFY

In ADF Studio, click Test Connection on your Linked Service. If it fails, use the error table below to diagnose.

Error Code	Cause	Fix
AADSTS700027	Cert in ADF doesn't match App Registration	Re-export .cer and re-upload to Azure AD App Registration
Invalid certificate	Base64 string corrupted or truncated	Re-run PowerShell export, copy entire content of cert_base64.txt
403 Forbidden	Graph API permissions missing	Add required permissions and click Grant Admin Consent
Decryption failed	Wrong PFX password in Linked Service	Ensure password matches the one used in Export-PfxCertificate
HasPrivateKey=False	Cert installed on wrong machine	Run certreq -accept on the same machine that ran certreq -new

STEP 9 — CERTIFICATE RENEWAL PROCESS

Certificates expire. When renewing, only the certificate changes — your App Registration permissions remain completely untouched.

Check Expiry Date

```
# List all certs and expiry dates
Get-ChildItem -Path 'Cert:\LocalMachine\My' |
  Select-Object Subject, Thumbprint, NotAfter |
  Sort-Object NotAfter

# Alert if expiring within 30 days
Get-ChildItem 'Cert:\LocalMachine\My' | Where-Object {
  $_.NotAfter -lt (Get-Date).AddDays(30)
} | Select-Object Subject, NotAfter
```

What Changes vs What Stays the Same on Renewal

Item	On Renewal
Generate new CSR	Yes — repeat Step 1
Send CSR to Cyber team	Yes — repeat Step 2
Install new cert	Yes — certreq -accept
Export new PFX + CER	Yes — repeat Step 3
Upload new CER to Azure AD	Yes — remove old, upload new
Update Base64 in ADF Linked Service	Yes — paste new Base64 string
API Permissions	NO — never changes with cert rotation
Admin Consent	NO — does not need re-granting
Client ID / Tenant ID	NO — stays the same
Private Key (if same machine)	Can reuse existing or generate new

QUICK REFERENCE CHEAT SHEET

```
# ■ STEP 1: Generate CSR ████████████████████████████████████████████████████████  
certreq -new 'C:\certs\request.inf' 'C:\certs\request.csr'  
  
# → Send request.csr to Cyber Security team  
  
# ■ STEP 2: Install Signed Cert (after Cyber team returns .cer) ■■  
certreq -accept 'C:\certs\mycert.cer'  
  
# ■ STEP 3: Verify installation ████████████████████████████████████████████████████████  
Get-ChildItem 'Cert:\LocalMachine\My' | Where-Object {  
    $_.Subject -like '*ADF-GraphAPI*' } | Select HasPrivateKey  
  
# ■ STEP 4: Export PFX + CER ████████████████████████████████████████████████████████  
$cert = Get-ChildItem 'Cert:\LocalMachine\My' | Where-Object {  
    $_.Subject -like '*ADF-GraphAPI*' }  
$pwd = ConvertTo-SecureString 'YourPassword123' -Force -AsPlainText  
Export-PfxCertificate -Cert $cert -FilePath 'myapp.pfx' -Password $pwd  
Export-Certificate -Cert $cert -FilePath 'myapp.cer' -Type CERT  
  
# ■ STEP 5: Convert to Base64 ████████████████████████████████████████████████████████  
$bytes = [System.IO.File]::ReadAllBytes('myapp.pfx')  
$base64 = [System.Convert]::ToBase64String($bytes)  
$base64 | Out-File 'cert_base64.txt' -Encoding ASCII  
  
# ■ STEP 6: Check expiry ████████████████████████████████████████████████████████  
Get-ChildItem 'Cert:\LocalMachine\My' | Select Subject, NotAfter
```

Key Principles to Remember

- Certificate = Authentication only (proves who you are). Permissions = Authorization (what you can do).
- The private key never leaves your machine — only the CSR and CER are shared externally.
- Exportable = TRUE in request.inf is mandatory — without it you cannot export the PFX for ADF.
- certreq -accept must run on the same machine that ran certreq -new — this links the cert with the private key.
- Replacing or renewing a certificate never affects API permissions or Admin Consent.
- ADF uses the PFX (as Base64) every time it authenticates — the private key signs each JWT assertion sent to Azure AD.

Files Summary

File	Contains	Who Needs It	Safe to Share?
request.inf	CSR config	You only	Yes
request.csr	Public key + identity	Send to Cyber team	Yes
mycert.cer (from Cyber team)	Signed public cert	You + Azure AD upload	Yes
myapp.pfx	Private key + cert	ADF (via Base64)	NO — keep secret
myapp.cer	Public cert only	Azure AD App Reg	Yes
cert_base64.txt	Base64 encoded PFX	Paste into ADF	NO — contains private key